Can Synthetic Spam Beat Real-World Detectors? Evaluating LLMs' Dual Role in Spam Generation and Detection

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Abstract: Large Language Models (LLMs) fundamentally reshape spam detection by enabling both synthetic training data generation and AI-powered spam attacks. We systematically evaluate this dual impact through comprehensive experimental tracks covering synthetic-to-real training scenarios, zero-shot detection of AI-generated attacks, and cross-model augmentation strategies. Our rigorous evaluation across multiple LLMs, prompt variations, and classifier architectures reveals three critical findings. First, synthetic spam can partially substitute for real training data with dramatic model-dependent quality gaps: high-quality generation maintains 86% of baseline performance while low-quality generation retains only 54%. Second, zero-shot detection of AI-generated spam achieves moderate baseline performance, with substantial variation across generation sources. Third, cross-model synthetic augmentation significantly improves detection effectiveness, with diversity-driven training providing larger gains than quality-optimized approaches. Ensemble classifiers consistently demonstrate superior robustness to distributional shifts compared to linear models. These findings reveal asymmetric advantages favoring attackers through accessible generation capabilities, though defenders can leverage robust architectures and cross-model augmentation to maintain detection effectiveness. Our work provides empirical foundations for understanding Generative AI's role in cybersecurity offense-defense dynamics.

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1 INTRODUCTION

Generative AI technologies have improved productivity across numerous domains, accelerating knowledge dissemination and democratizing access to sophisticated content creation capabilities (Baldassarre et al., 2023). However, this technological advancement has introduced profound challenges to cybersecurity, fundamentally altering both offensive and defensive capabilities (Wach et al., 2023). The dual-use nature of these technologies presents a critical question for security practitioners (Korimilli et al., 2025; Mirtaheric et al., 2025): do generative models ultimately favor attackers or defenders?

This question carries particular urgency in the domain of spam and phishing detection. Adversaries increasingly exploit Large Language Models (LLMs) to generate deceptive content at an unprecedented scale (Bethany et al., 2024; Weinz et al., 2025). Unlike template-based generation methods that dominated previous eras, LLM-generated spam exhibits natural linguistic characteristics, grammatical correctness, and contextual coherence that closely approximate legitimate communications. Eliminating traditional indicators such as spelling errors, grammatical errors, and repetitive patterns fundamentally undermines detection heuristics developed over decades of adversarial evolution (Magdy et al., 2022). Furthermore, LLMs enable sophisticated personalization strategies through integration with information gathering capabilities (Yu et al., 2025; Kim et al., 2025), allowing adversaries to craft targeted spear phishing campaigns informed by publicly available intelligence on potential victims. This convergence of generation quality, operational scale, and personalization capacity represents a qualitative shift in the threat landscape.

Defenders confronting this evolving threat face their own fundamental challenges. Spam detection systems require large, diverse training datasets, yet acquisition of such data encounters critical obstacles including privacy regulations, legal constraints, and

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the inherent rarity of malicious samples in operational environments (Bazzaz Abkenar et al., 2021). Traditional machine learning approaches to spam classification have demonstrated effectiveness across multiple paradigms (Wang et al., 2025), but all depend critically on training data that reflects the distributional characteristics of deployment environments. The same LLM technologies that empower attackers also present defenders with potential solutions to data scarcity through synthetic generation (Zhang et al., 2025). If synthetic spam can effectively substitute for real training data without degrading detection performance, defenders gain transformative capabilities for building detection systems without accessing sensitive data and rapidly adapting to emerging threats.

The central challenge, however, extends beyond training data augmentation. Organizations must defend against adversaries who may employ any of numerous capable language models for attack generation, yet defenders typically cannot predict which specific models adversaries will utilize. This asymmetry raises critical questions about cross-model generalization: can detection systems trained on synthetic data from accessible LLMs effectively recognize attacks generated by different LLM sources? If cross-model augmentation proves effective, defenders could proactively strengthen their systems by generating synthetic training data from available models, effectively leveraging generative AI to counter generative AI-powered attacks.

These considerations motivate three complementary research questions that systematically examine whether Generative AI favors attackers or defenders in spam detection:

1. **RQ1: Can LLM-generated synthetic spam effectively substitute for real training data?**
2. **RQ2a: Can traditional spam detectors recognize LLM-generated attacks without prior exposure?**
3. **RQ2b: Does augmenting training data with synthetic spam from available LLMs improve detection of attacks from different LLM sources?**

This work makes several fundamental contributions to understanding Generative AI's role in spam detection:

1. **Dual-Perspective Evaluation Framework.** We introduce a systematic framework for evaluating LLMs' dual impact on spam detection, explicitly addressing both offensive capabilities for attack generation and defensive capabilities for training augmentation. This approach provides the first comprehensive assessment of whether generative AI fundamentally favors attackers or defenders.
2. **Comprehensive Cross-Model Analysis.** We establish a rigorous methodology for evaluating cross-model generalization dynamics, examining whether synthetic data from one LLM source improves detection of attacks from different LLM sources. This validation framework addresses the critical operational scenario where defenders and adversaries employ different models.
3. **Robustness and Architectural Sensitivity Analysis.** We systematically characterize how classifier architecture affects synthetic data utility, providing evidence that architectural selection constitutes a critical defensive decision. Our analysis reveals fundamental differences in how various classifier families respond to distributional shifts introduced by synthetic augmentation.
4. **Statistical Methodology for Synthetic Data Evaluation.** We contribute replicated experimental designs and statistical frameworks that quantify uncertainty, effect magnitudes, and performance degradation patterns. Our regression models enable practitioners to predict detection performance across arbitrary synthetic-to-real mixing ratios and establish evidence-based deployment boundaries.
5. **Public Benchmark Dataset.** We release a comprehensive benchmark comprising LLM-generated spam samples across multiple models and prompt strategies, enabling reproducibility and standardized evaluation for future research[1].

2 RELATED WORK

Spam and phishing email detection has evolved through multiple paradigm shifts driven by adversarial co-evolution between attackers and defenders. Early systems relied on rule-based filtering and manually curated blacklists (Al-Kabbi et al., 2023; Magdy et al., 2022), which proved brittle against adaptive attackers. The field subsequently transitioned to machine learning approaches, with statistical classifiers including Naive Bayes, Support Vector Machines, and ensemble methods (Kontsewaya et al., 2021; Wijaya et al., 2023; Budiman et al., 2024; Bazzaz Abkenar et al., 2021; Wang et al., 2025) demonstrating superior generalization through learned feature representations. Recent advances in deep learning have further improved detection capabilities, with

[1]https://github.com/tisage/llm-synthetic-spam-benchmark

transformer-based models leveraging contextualized embeddings (Altwaijry et al., 2024; Thakur et al., 2023).

However, two persistent challenges limit practical deployment effectiveness. First, severe class imbalance characterizes operational environments where malicious emails constitute only 1-10% of total traffic (Bazzaz Abkenar et al., 2021; Johari et al., 2025), causing classifiers to develop strong bias toward majority classes and sacrificing minority class sensitivity (Wang et al., 2025). Second, privacy regulations and legal constraints restrict retention and sharing of email data, limiting access to large-scale training datasets (Kim et al., 2023). These dual challenges motivate exploration of synthetic data generation as a potential solution.

Synthetic data generation addresses training data scarcity by creating artificial examples that augment limited real datasets. Traditional approaches in natural language processing have explored augmentation techniques including synonym replacement, back-translation, and paraphrasing (Shorten et al., 2021), though these methods lack semantic understanding and cannot generate content with controlled characteristics while maintaining realistic coherence. Deep generative models such as Generative Adversarial Networks have shown effectiveness for structured cybersecurity data including network traffic and intrusion detection (Bourou et al., 2021; Ammara et al., 2025; Zouhri and Idri, 2024; Agrawal et al., 2024), but face critical limitations for text generation due to difficulties maintaining semantic coherence and linguistic realism required for spam generation.

Large Language Models(LLMs) have fundamentally transformed synthetic text generation for spam and phishing attacks through pre-training on massive corpora spanning diverse linguistic patterns. Unlike feature-space interpolation or adversarial training, LLMs generate text through autoregressive prediction conditioned on context, enabling semantically coherent content that maintains consistency across multiple attributes (Razmyslovich et al., 2025; Almorjan et al., 2025). This makes LLMs the preferred choice for attackers requiring rapid, large-scale, cost-effective spam generation through simple API calls without model training. Recent applications to cybersecurity have explored domain-driven synthetic text generation for classification tasks (Razmyslovich et al., 2025; Zhang et al., 2025), structured data creation incorporating domain knowledge (Almorjan et al., 2025), and general cyber defense applications (Ammara et al., 2024). Systematic reviews analyzing hundreds of LLM applications (Zhang et al., 2025; Ding et al., 2023) document broad adoption across security domains, while emerging research explores prompt engineering strategies for controlled generation (Chen et al., 2025; de Araujo-Filho et al., 2023).

However, existing work exhibits critical limitations. Prior evaluations focus predominantly on single-sided assessments either defensive utility for classifier training or offensive threats for spam generation without systematically characterizing the dual-use nature of LLMs where the same generation capabilities empower both attackers and defenders. Recent work has demonstrated LLMs' potential for generating convincing phishing emails and automating social engineering attacks (Bethany et al., 2024; Afane et al., 2024; Weinz et al., 2025; Ai et al., 2024; Thapa et al., 2025), yet comprehensive evaluation frameworks simultaneously assessing both offensive threats and defensive countermeasures under controlled experimental conditions remain absent. Furthermore, cross-model generalization dynamics that whether training on synthetic data from one LLM improves detection of spam from different LLM sources remain unexplored despite operational importance when attackers and defenders employ different models. Rigorous experimental methodology employing replicated trials and systematic exploration of generation parameters is largely absent from prior work.

3 METHODOLOGY

We propose a systematic evaluation framework to assess whether generative AI fundamentally favors attackers or defenders in spam detection. Our framework consists of five principal components: data preparation through stratified sampling, synthetic spam generation with varied LLM prompt strategies, systematic mixing of real and synthetic training data, classification, and comprehensive evaluation with rigorous statistical analysis. This design enables assessment of synthetic data utility across different operational scenarios while ensuring statistical rigor and reproducible conclusions.

3.1 Data Sampling Design

We partition the spam corpus into R independent groups using stratified random sampling, with each group containing N samples at fixed spam-to-ham ratio ρ. Sampling without replacement ensures statistical independence between groups, satisfying assumptions for parametric hypothesis testing. This replicated design enables calculation of performance variance across groups, construction of confidence inter-

vals, and rigorous hypothesis testing with quantified uncertainty rather than point estimates.

To eliminate test set variance as a confounding factor, we maintain a fixed held-out test set across all experimental conditions. The test set remains constant while training sets vary according to synthetic proportions, ensuring that observed performance differences reflect genuine treatment effects rather than evaluation artifacts from test set composition variability. Specific values for R, N, and ρ are determined through power analysis and specified in Section 4.

3.2 Synthetic Data Generation

We employ Large Language Models to generate synthetic spam through prompt-guided text generation. Given a real spam email as a seed, the LLM produces a synthetic variant maintaining semantic similarity while introducing lexical and structural variation. We design three systematically varied prompt strategies to explore how generation instructions influence synthetic data quality:

- **Original:** Faithfully reproduces spam characteristics while introducing natural lexical variation, maintaining semantic similarity and malicious intent without amplification or attenuation.
- **Strong:** Explicitly amplifies spam indicators by emphasizing urgency signals, persuasive elements, and social engineering techniques.
- **Weak:** Produces subtle spam with reduced explicit indicators, generating content that appears more legitimate while retaining underlying spam nature.

This controlled variation enables systematic investigation of whether prompt intensity affects generation quality and classifier performance. Complete prompt templates are specified in Section 4 and Appendix.

3.3 Cross-Group Mixing Strategy

We employ cross-group mixing where training data from group i combines with synthetic data generated from different groups $j \neq i$. This design deliberately introduces distributional mismatch between generation sources and training targets, reflecting realistic deployment scenarios where exact distribution matching between synthetic generation sources and operational training data is infeasible or unverifiable.

Cross-group mixing provides two critical advantages over within-group alternatives. First, it tests whether synthetic data provides value through semantic diversity rather than precise distribution matching, establishing whether LLM-generated spam captures generalizable features beyond specific distributional artifacts. Second, it better reflects operational constraints, including temporal flexibility (using historical synthetic data for current threats), source diversity (pooling synthetic data from multiple corpora), and organizational collaboration (sharing synthetic samples without revealing operational distributions). Our preliminary analysis demonstrated that cross-group mixing maintains comparable performance to within-group mixing while providing more conservative estimates of synthetic data utility under realistic deployment conditions[2].

3.4 Evaluation Framework: Three Experimental Tracks

Our evaluation framework encompasses three complementary tracks that systematically address the research questions outlined in Section 1.

Track 1: Synthetic-to-Real Detection (RQ1): This track evaluates whether LLM-generated synthetic spam can effectively train classifiers to detect real spam. Classifiers train on mixtures of real ham (fixed) and varying proportions of real spam combined with synthetic spam, then test on identical real-world data. As the synthetic proportion θ varies from 0 (pure real) to 1 (pure synthetic), we quantify performance degradation patterns, establishing trade-offs between reduced real data requirements and maintained detection accuracy. The test set remains purely real throughout, ensuring results reflect operational deployment scenarios where detectors must identify genuine spam.

Track 2a: Real-to-Synthetic Detection (RQ2a): This track evaluates whether classifiers trained solely on real spam can detect LLM-generated synthetic spam, representing zero-shot detection scenarios. Training employs only real ham and real spam without synthetic exposure, while testing evaluates the detection of LLM-generated synthetic spam. Strong baseline performance would indicate that features distinguishing real spam from ham generalize to identifying AI-generated spam, suggesting inherent robustness. Weak baseline performance would indicate that AI-generated spam represents a novel threat requiring adaptive training strategies.

Track 2b: Mixed-to-Synthetic Detection (RQ2b): This track investigates whether incorporating synthetic data from other LLMs improves

[2]Cross-group strategy shows consistent advantages over within-group strategy: mean F1 improvement of +0.009 ($p < 10^{-6}$, Cohen's $d = 0.13$) across all configurations.

detection capabilities through cross-model augmentation. Classifiers train on real ham plus varying mixtures of real spam and synthetic spam from LLM B, then test on detecting synthetic spam from LLM A (different from training). This cross-model design evaluates whether features learned from one LLM's synthetic patterns generalize to detecting another LLM's spam. Performance improvement over Track 2a baseline would demonstrate that cross-model synthetic augmentation enhances detection capabilities, enabling "fighting fire with fire" defensive strategies.

4 EXPERIMENTAL CONFIGURATION

This section specifies the complete experimental configuration required to reproduce our evaluation framework. Following the methodology established in Section 3, we provide concrete parameter values for dataset composition, LLM configurations, experimental tracks, classification settings and statistical analysis procedures. Detailed hyperparameter grids, prompt templates, and LLM configurations are provided in Appendix Table 8.

4.1 Dataset and Sampling

We employ the CEAS-08 spam email dataset (Champa et al., 2024), comprising 39,154 real-world email samples (21,842 spam, 17,312 legitimate) available in our repository[1]. The corpus is partitioned into $R = 20$ independent groups through stratified random sampling, with each group containing $N = 1{,}000$ samples at a fixed spam ratio $\rho = 0.1$ (100 spam, 900 ham). This 10% imbalance reflects realistic operational conditions. The dataset is clean and structured. An 80-20 train-test split establishes a fixed held-out test set held constant across all experiments, eliminating test set variance as a confounding factor.

4.2 LLM Models and Generation

We employ two state-of-the-art large language models: *GPT-4.1-mini* (OpenAI) and *Claude-3.5-Haiku* (Anthropic). These models offer optimal cost-performance trade-offs for realistic spam generation scenarios, providing sufficient context windows while maintaining low per-token costs economically viable for large-scale campaigns. Both models employ identical generation parameters: temperature = 0.7, max tokens = 1,000, top-p = 1.0. Complete API configurations, cost analysis, and parameter justifications are provided in Appendix Table 5.

We implement three systematically varied prompt strategies to explore generation intensity: **Original** (faithful reproduction with lexical variation), **Strong** (amplified spam indicators), and **Weak** (subtle spam with reduced explicit signals). Each prompt follows a consistent JSON-formatted template ensuring reliable parsing. Complete prompt templates are provided in supplementary materials to support experimental reproducibility. Table 1 illustrates how prompt variation affects generation quality across the three strategies.[3]

Prompt Template Structure:
`You are an expert email content analyzer. Given the following spam email, generate a semantically similar but lexically distinct spam variant.`
[Strategy-Specific Instructions]
`Original Email:` { **original_spam_content** }
`Output Format (JSON):`
{`"subject": "...", "body": "..."`}

4.3 Experimental Tracks

Following the methodology framework (Section 3), we implement three experimental tracks with specific parameter configurations. All tracks employ cross-group mixing, no synthetic exposure, all three prompt strategies, and 20 independent replications. Table 2 summarizes the key configuration differences.

Track 1 (RQ1): Varies synthetic proportion θ from 0% to 100% in 10% increments to quantify performance degradation as real spam is replaced by synthetic spam during training. Training sets contain $(1-\theta) \times 100$ real spam + $\theta \times 100$ synthetic spam + 900 ham. Testing employs 100 real spam + 900 ham (fixed across all ratios).

Track 2a (RQ2a): Varies training data size to evaluate whether additional real spam improves zero-shot detection of LLM-generated synthetic spam. Training employs 100%, 150%, or 200% real spam (100, 150, or 200 samples) with 900 ham. Testing evaluates detection of LLM-generated synthetic spam across both models and all prompt strategies.

Track 2b (RQ2b): Evaluates cross-model augmentation effectiveness at three synthetic ratios ($\theta = 0\%, 50\%, 100\%$). Training employs synthetic spam

[3] *Ethics Note:* All synthetic spam emails are generated solely for research purposes to advance defensive cybersecurity capabilities. Complete prompt templates are provided in supplementary materials to support reproducibility.

Table 1: Representative Synthetic Spam Examples (GPT-4.1-mini). Original prompts produce natural spam with realistic language patterns, strong prompts amplify urgency cues and explicit offers, while weak prompts generate subtle spam with professional tone.

Strategy	Generated Email
Original	*Subject:* Congratulations! You've Won $5.5 Million *Body:* Dear Winner, Your email has been randomly selected in our international lottery draw. You have won $5,500,000.00! To claim your prize, contact our claims department immediately with your personal details. Act now - this offer expires in 48 hours! http://prize...
Strong	*Subject:* URGENT: $5.5M Lottery Prize - CLAIM NOW! *Body:* IMMEDIATE ACTION REQUIRED! You are the GUARANTEED winner of $5.5 MILLION DOLLARS! This is a LIMITED TIME OFFER - only 24 hours left! Don't let this life-changing opportunity slip away... Contact us RIGHT NOW or lose your prize forever! http://urgent...
Weak	*Subject:* Notification Regarding Prize Selection *Body:* Dear Recipient, We are pleased to inform you that your email address was included in our recent prize selection process. You may be eligible to receive a substantial financial award. For verification and processing, please provide your contact information at your earliest convenience... http://verify...

Table 2: Track-Specific Parameters.

Parameter	Track 1	Track 2a	Track 2b
Training spam	Real + Synthetic	Real only	Real + Synthetic
Testing spam	Real	Synthetic	Synthetic
Controlled variable	Synthetic ratio θ	Training size	Synthetic ratio θ
Variable values	0%, 10%, ..., 100%	100%, 150%, 200%	0%, 50%, 100%

from one LLM while testing evaluates detection of synthetic spam from the opposite LLM. For example, training on Claude synthetic spam tests detection of GPT synthetic spam.

4.4 Classification Configurations

We employ two classifiers representing distinct learning paradigms: Support Vector Machine (SVM) and Random Forest (RF). All emails are transformed to TF-IDF feature vectors, with the vectorizer fitted exclusively on training data to prevent information leakage. Hyperparameters are optimized through grid search with 5-fold stratified cross-validation using F1-score as the optimization metric. Complete hyperparameter grids and TF-IDF configurations are provided in Appendix Table 6

4.5 Evaluation Metrics

We compute descriptive statistics (mean, standard deviation) across 20 independent groups for each configuration. Paired *t*-tests compare each synthetic ratio against baseline with Benjamini-Hochberg FDR correction ($\alpha = 0.05$) for multiple comparisons. Cohen's *d* effect sizes quantify practical importance beyond statistical significance. Linear regression models performance degradation as a function of synthetic proportion. F1-score serves as the primary metric, with supporting metrics including Accuracy, Precision, Recall, and AUC-ROC.

The complete evaluation framework comprises 4,080 training-evaluation units across all tracks (Track 1: 2,640; Track 2a: 720; Track 2b: 720), covering all combinations of LLM models, prompt strategies, classifiers, and replications. All experiments employ official LLM APIs (GPT-4.1-mini, Claude-3.5-Haiku) with total 12,000 synthetic samples. Classifier training executes on standard computing infrastructure (Intel Xeon 32-core CPU, 64 GB RAM). Complete cost breakdown, infrastructure specifications, and experimental results are provided in Appendix Table 5 and our repository[1].

5 EVALUATION

Our evaluation systematically assesses LLM-generated synthetic spam across three complementary experimental tracks spanning 4,080 training-evaluation units. Following the methodology framework, we present results organized by research questions: Track 1 evaluates synthetic data for classifier training (RQ1), Track 2a establishes baseline detection capabilities against AI-powered spam (RQ2a), and Track 2b investigates hybrid training strategies (RQ2b).

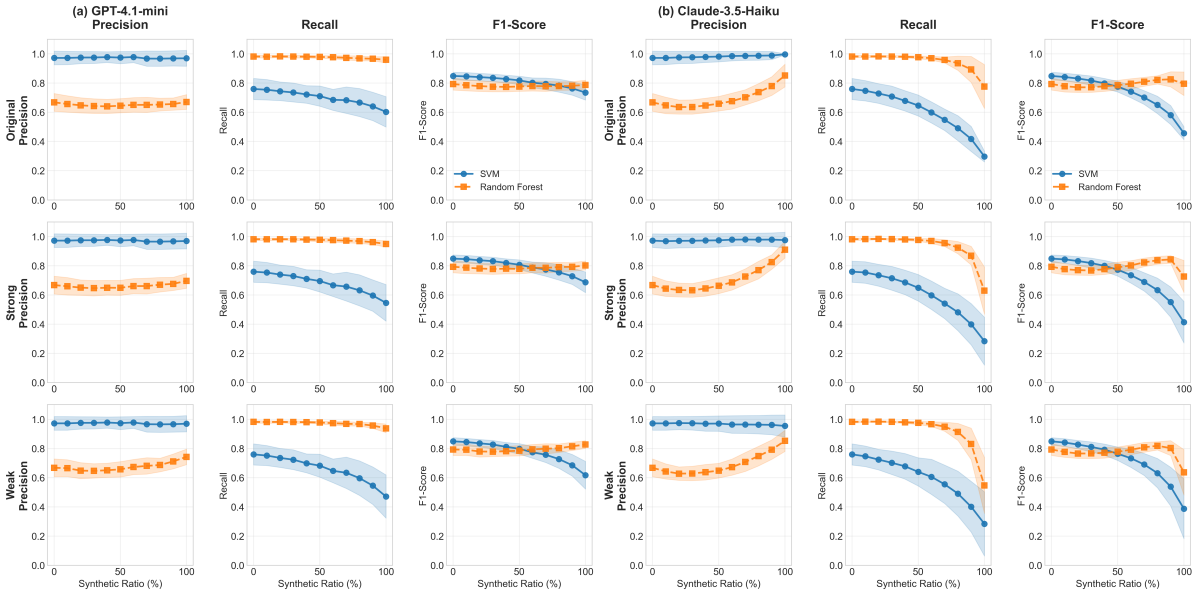


Figure 1: Track 1 - Synthetic-to-Real Detection: Performance metrics (Precision, Recall, F1-Score) across prompt strategies (original/strong/weak, rows) and models (columns). (a) GPT: F1-score exhibits prompt invariance with nearly identical degradation (0.85 $\rightarrow$ 0.70-0.74 at 100% synthetic, -13.4% relative). SVM maintains high precision but suffers recall loss. (b) Claude: Shows prompt sensitivity with original/strong prompts degrading moderately (F1 $\rightarrow$ 0.45-0.55 at 100%, -46.2% relative), while weak prompt exhibits severe degradation (F1 $\rightarrow$ 0.30-0.40). Shaded regions: $\pm$1 SD across 20 groups.

Table 3: Track 1: Synthetic-to-Real Detection Performance - SVM and Random Forest (RF) Comparison.

Model	Classifier	Prompt	Baseline (0% Synthetic)			100% Synthetic		
			Precision	Recall	F1	Precision	Recall	F1
GPT-4.1-mini	SVM	Original	0.972 ± 0.047	0.759 ± 0.072	0.849 ± 0.026	0.970 ± 0.054	0.603 ± 0.104	0.735 ± 0.051
		Strong				0.970 ± 0.054	0.546 ± 0.125	0.687 ± 0.071
		Weak				0.969 ± 0.056	0.471 ± 0.148	0.617 ± 0.095
	Random Forest	Original	0.668 ± 0.061	0.982 ± 0.008	0.793 ± 0.041	0.670 ± 0.051	0.959 ± 0.022	0.788 ± 0.032
		Strong				0.697 ± 0.051	0.949 ± 0.021	0.802 ± 0.030
		Weak				0.743 ± 0.053	0.938 ± 0.029	0.827 ± 0.026
Claude-3.5-haiku	SVM	Original	0.972 ± 0.047	0.759 ± 0.072	0.849 ± 0.026	0.997 ± 0.004	0.297 ± 0.037	0.457 ± 0.045
		Strong				0.976 ± 0.054	0.284 ± 0.164	0.414 ± 0.142
		Weak				0.954 ± 0.075	0.284 ± 0.220	0.387 ± 0.205
	Random Forest	Original	0.668 ± 0.061	0.982 ± 0.008	0.793 ± 0.041	0.852 ± 0.078	0.776 ± 0.150	0.796 ± 0.081
		Strong				0.911 ± 0.057	0.630 ± 0.162	0.727 ± 0.110
		Weak				0.852 ± 0.072	0.547 ± 0.192	0.637 ± 0.156

5.1 Track 1: Synthetic-to-Real Detection

As Figure 1 and Table 3 show, performance degradation follows highly linear patterns with increasing synthetic proportion, yielding $R^2 > 0.95$ for SVM configurations (Appendix Table 7). Random Forest exhibits near-zero slopes across most configurations, indicating stable performance regardless of synthetic proportion.

Classifier architecture determines robustness to synthetic training data. Random Forest demonstrates substantially higher resilience than SVM across both generation models. For GPT at 100% synthetic training, RF maintains near-baseline F1 (0.788-0.827 vs baseline 0.793), while SVM degrades significantly (0.617-0.735 vs baseline 0.849). For Claude original prompt, RF achieves F1 = 0.796 compared to SVM's 0.457. This architectural difference manifests in precision-recall trade-offs: SVM maintains near-perfect precision (0.95-1.00) while recall degrades substantially, resulting in overly conservative classification. RF maintains balanced profiles with both metrics stable under synthetic substitution, suggesting its ensemble voting mechanism provides more robust generalization than SVM's strict decision boundaries.

Prompt strategy affects effectiveness differently across classifiers. For SVM, original prompts consistently outperform alternatives across both models, establishing that faithful reproduction produces

synthetic spam closest to real spam's feature distribution. RF exhibits non-monotonic behavior with prompt sensitivity varying by model.

GPT generates higher-quality synthetic training data than Claude across all configurations. At 100% synthetic with original prompts, GPT maintains 86.6% of baseline performance while Claude retains only 53.8%. This quality gap persists across all prompt strategies and classifiers, indicating fundamental differences in generation capabilities.

5.2 Track 2a: Real-to-Synthetic Detection

As Figure 2 and Table 4 show, classifier architecture determines sensitivity to training data quantity for zero-shot detection. SVM exhibits substantial monotonic improvement with additional real spam (GPT detection: F1 from 0.744 at 100 samples to 0.881 at 200 samples), while Random Forest maintains stable performance across training sizes (F1 $\approx$ 0.83 for GPT, $\approx$ 0.80 for Claude). This reflects architectural differences between margin-based and ensemble learning.

GPT synthetic spam exhibits higher baseline detectability (SVM: 0.636-0.744, RF: 0.830-0.831) than Claude synthetic spam (SVM: 0.335-0.589, RF: 0.747-0.811) at 100 training samples. This higher detectability paradoxically correlates with GPT's superior performance in Track 1, suggesting distributional proximity to real spam enables both effective classifier training and detection. Weak prompts universally yield lowest detection (Claude weak: SVM F1 = 0.335), indicating prompt optimization for evasiveness differs by LLM source.

5.3 Track 2b: Mixed-to-Synthetic Detection

As Figure 3 and Table 4 show, training with synthetic spam from one LLM improves detection of another LLM's synthetic spam. For Claude detection using GPT synthetic training data, SVM achieves substantial improvements at 100% synthetic (strong prompt: F1 from 0.590 to 0.860). For GPT detection using Claude synthetic training data, RF achieves F1 improvement from 0.824 to 0.894 for original prompt. Cross-model augmentation provides measurable benefits across configurations.

Augmentation effectiveness depends on prompt diversity rather than quality. For Claude detection, weak prompt training data achieves F1 = 0.805 compared to original prompt training (F1 = 0.688), suggesting diverse, lower-quality synthetic spam better captures the feature space of AI-generated content than high-quality samples alone.

Random Forest demonstrates superior cross-model robustness compared to SVM. For both GPT and Claude detection, RF maintains stable baseline performance across prompts (F1 $\approx$ 0.75-0.82) and achieves consistent improvements with synthetic augmentation. SVM exhibits substantial baseline variance (F1 range: 0.335-0.744) and less consistent augmentation benefits, indicating ensemble methods extract cross-model invariant features more effectively.

Optimal synthetic ratios vary by target model and classifier. For Claude detection using GPT synthetic training data, most configurations show monotonic improvement. For GPT detection using Claude synthetic training data, RF benefits from high synthetic ratios (optimal at 100%, F1 = 0.834-0.908), while SVM demonstrates non-monotonic behavior with peak performance at 50% before declining at 100%. This suggests SVM overfits to distributional artifacts when trained exclusively on cross-model synthetic spam.

6 DISCUSSION

Our experiments reveal three fundamental patterns across all evaluation tracks. First, model-dependent performance asymmetry persists throughout: GPT synthetic data maintains superior training effectiveness while exhibiting higher detectability, whereas Claude shows reduced utility and detectability. Second, ensemble architectures demonstrate consistent superiority over margin-based methods. Third, cross-model synthetic augmentation provides measurable detection improvements, with magnitudes varying by baseline performance, prompt strategy, and classifier architecture.

GPT's superior generation quality stems from alignment to real spam patterns, preserving lexical diversity and statistical regularities that enable effective classifier training. However, this proximity creates a fundamental paradox: the same fidelity that makes GPT synthetic spam valuable for training also renders it more detectable when classifiers are trained on real spam. Claude exhibits greater divergence, simultaneously reducing both training utility and detectability. This suggests high-quality synthetic generation inherently requires balancing fidelity for augmentation against divergence for evasion.

Random Forest's consistent superiority stems from bootstrap aggregation's resistance to distributional shifts. Multiple decision trees trained on random feature and sample subsets capture diverse

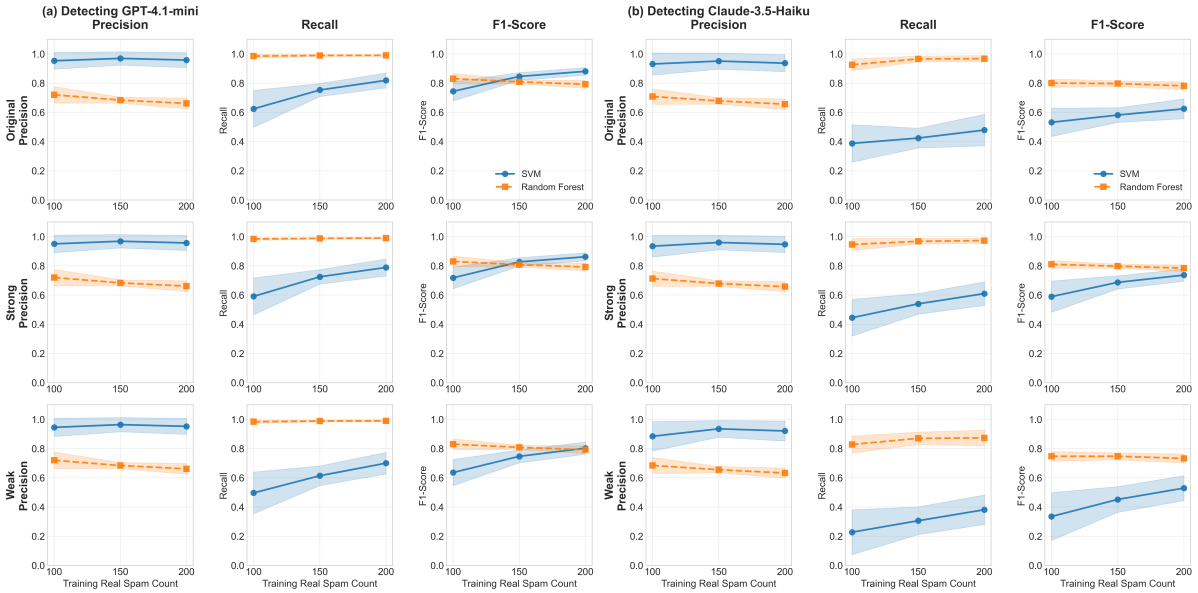


Figure 2: Track 2a - Real-to-Synthetic Detection: Performance metrics (Precision, Recall, F1-Score) across prompt strategies (rows) and models (columns). (a) Detecting GPT spam: SVM F1 = 0.74-0.82, RF F1 = 0.82-0.87, with SVM showing increasing F1 as training data grows while RF remains stable. (b) Detecting Claude spam: SVM shows moderate detection improving with more training data (F1 = 0.53-0.74 across prompts), while RF maintains strong performance (F1 = 0.80-0.85).

Table 4: Track 2a & 2b: Real-to-Synthetic and Mixed-to-Synthetic Detection Performance.

Testing	**Prompt**	**Classifier**	**Track 2a: Real-to-Synthetic**			**Track 2b: Mixed-to-Synthetic**		
Target			**100%**	**150%**	**200%**	**0%**	**50%**	**100%**
GPT-4.1-mini	Original	SVM	0.744 ± 0.064	0.846 ± 0.025	0.881 ± 0.025	0.744 ± 0.064	0.770 ± 0.035	0.649 ± 0.042
		Random Forest	0.831 ± 0.035	0.809 ± 0.014	0.792 ± 0.026	0.824 ± 0.025	0.816 ± 0.023	0.894 ± 0.030
	Strong	SVM	0.718 ± 0.072	0.827 ± 0.028	0.862 ± 0.027	0.718 ± 0.072	0.822 ± 0.040	0.696 ± 0.110
		Random Forest	0.830 ± 0.035	0.808 ± 0.014	0.792 ± 0.026	0.824 ± 0.026	0.827 ± 0.032	0.908 ± 0.040
	Weak	SVM	0.636 ± 0.087	0.747 ± 0.043	0.803 ± 0.041	0.636 ± 0.086	0.822 ± 0.038	0.688 ± 0.119
		Random Forest	0.830 ± 0.036	0.808 ± 0.014	0.792 ± 0.026	0.824 ± 0.028	0.822 ± 0.026	0.834 ± 0.061
Claude-3.5-haiku	Original	SVM	0.532 ± 0.095	0.582 ± 0.049	0.624 ± 0.066	0.532 ± 0.094	0.666 ± 0.031	0.688 ± 0.033
		Random Forest	0.800 ± 0.027	0.797 ± 0.018	0.781 ± 0.026	0.802 ± 0.021	0.802 ± 0.020	0.802 ± 0.022
	Strong	SVM	0.589 ± 0.104	0.687 ± 0.044	0.737 ± 0.040	0.590 ± 0.104	0.827 ± 0.032	0.860 ± 0.033
		Random Forest	0.811 ± 0.026	0.798 ± 0.013	0.784 ± 0.022	0.806 ± 0.023	0.814 ± 0.024	0.823 ± 0.027
	Weak	SVM	0.335 ± 0.160	0.452 ± 0.086	0.529 ± 0.083	0.335 ± 0.160	0.748 ± 0.038	0.805 ± 0.026
		Random Forest	0.747 ± 0.033	0.747 ± 0.023	0.732 ± 0.030	0.748 ± 0.031	0.777 ± 0.032	0.813 ± 0.021

patterns while avoiding overfitting to synthetic artifacts. Voting mechanisms average out sensitivities to anomalies, producing stable predictions across varying synthetic proportions. In contrast, SVM's margin-based learning optimizes single hyperplanes vulnerable to systematic biases, yielding linear degradation under same-model augmentation and non-monotonic behavior under cross-model training. Organizations should deploy Random Forest for stability, or SVM when predictable degradation patterns enable capacity planning.

Cross-model augmentation reveals that weak prompt training data provides larger detection improvements than strong prompts, demonstrating that diversity drives generalization more effectively than quality. This inverts optimization strategies: defenders should prioritize diverse, lower-quality data capturing broader feature spaces rather than training exclusively on high-quality samples.

LLMs fundamentally favor attackers through low cost, minimal technical barriers, and high effectiveness. Attackers can freely select optimal models and prompts for their objectives. Defenders face asymmetric constraints: effective augmentation requires cross-model data, ensemble architectures, and careful management of precision-recall trade-offs. Model quality gaps create advantages defenders cannot fully mitigate through architectural choices alone. Generative AI shifts the balance toward attackers unless defenders proactively adopt ensemble methods and cross-model augmentation strategies.

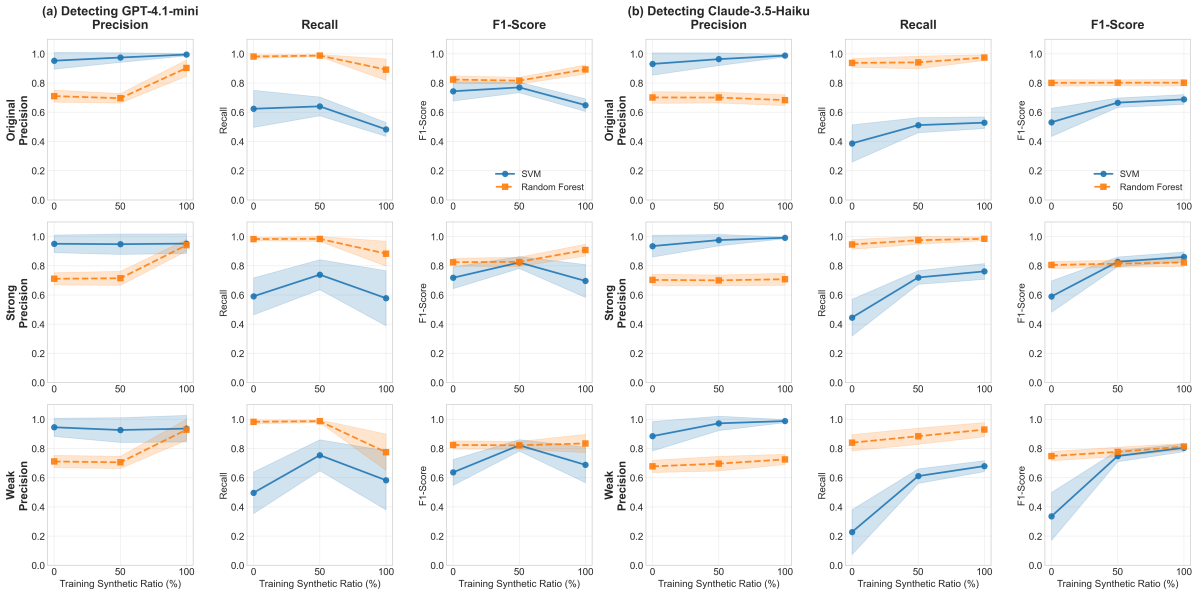


Figure 3: Track 2b - Mixed-to-Synthetic Detection: Performance metrics (Precision, Recall, F1-Score) across prompt strategies (original/strong/weak, rows) and detection targets (columns). (a) Detecting GPT (trained on Claude synthetic): SVM peaks at 50% synthetic (F1 = 0.77, +3.5%) before declining at 100% (F1 = 0.65, -13%). RF improves substantially with cross-model augmentation (F1: 0.82 $\rightarrow$ 0.89, +8.5%). (b) Detecting Claude (trained on GPT synthetic): SVM shows monotonic improvement (F1: 0.53 $\rightarrow$ 0.69, +30% relative). RF maintains stable high performance (F1 $\approx$ 0.80) across all ratios.

7 LIMITATIONS AND FUTURE WORK

Our evaluation acknowledges several limitations. TF-IDF captures lexical patterns but may miss semantic relationships that deep learning approaches could reveal. The CEAS-08 corpus represents 2008 spam characteristics; modern corpora may interact differently with LLM-generated content. Our fixed class imbalance represents one operational point; different imbalance ratios may exhibit different patterns. Our mid-tier models may not generalize to open-source LLMs, frontier models, or specialized approaches including fine-tuned models. Our cross-model detection focus leaves same-model scenarios unexplored. Spam detection findings may not generalize to other cybersecurity domains requiring different feature representations or threat models.

To address these limitations, future work will expand our evaluation in several directions. We plan to incorporate additional corpora including Nazario, Enron, and MeAJOR datasets, paired with deep learning classifiers such as Sentence-BERT models to capture semantic features beyond lexical patterns. Furthermore, we intend to integrate transformer-based architectures for comparative analysis, assessing their generalization capabilities across broader cybersecurity domains including social engineering detection and adversarial content generation. These extensions will provide a more comprehensive understanding of LLM detectability across diverse threat landscapes and architectural approaches.

8 CONCLUSION

This work systematically evaluates Generative AI's dual impact on spam detection, revealing fundamental asymmetries in offensive and defensive capabilities. Our findings demonstrate three critical insights: (1) LLM-generated synthetic spam can partially substitute real training data, though effectiveness varies dramatically by model generation quality; (2) AI-powered spam poses moderate detection challenges with performance strongly influenced by classifier architecture and training strategies; (3) cross-model synthetic augmentation significantly improves detection, with diversity-driven approaches outperforming quality-focused methods. Ensemble classifiers consistently demonstrate superior robustness to distributional shifts and cross-model generalization compared to linear models.

For cybersecurity practitioners, these results indicate that Generative AI currently favors attackers through accessible generation capabilities and low deployment costs. However, defenders can maintain

effectiveness by leveraging robust classifier architectures, cross-model training augmentation, and strategic synthetic data integration. Our statistical frameworks enable practitioners to predict detection performance across arbitrary synthetic mixing ratios, supporting evidence-based deployment decisions.

Future research should explore open-source LLMs to assess democratization effects, investigate transformer-based detection architectures, and evaluate generalization to broader cybersecurity domains including phishing, social engineering, and adversarial content generation.

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APPENDIX

This appendix provides detailed configuration parameters, hyperparameter settings, and statistical analysis supporting the experimental evaluation.

Table 5: LLM Configuration and API Costs.

Parameter	GPT-4.1-mini	Claude-3.5-Haiku
API endpoint	gpt-4.1-mini-2024-04-14	claude-3-5-haiku-20241022
Context	128K	200K
Input cost	$0.150 / 1M tokens	$0.250 / 1M tokens
Output cost	$0.600 / 1M tokens	$1.250 / 1M tokens

Table 6: Hyperparameters and TF-IDF Settings.

Classifier Hyperparameters (Grid Search)			
Classifier	**Parameter**	**Search Range**	**Optimal Value**
SVM	C	{0.1, 1.0, 10.0}	1.0
	Kernel	{linear, rbf}	linear
	Class weight	{None, balanced}	balanced
Random Forest	n_estimators	{50, 100, 200}	100
	max_depth	{5, 10, 20, None}	10
	min_samples_split	{2, 5}	5
	min_samples_leaf	{1, 2}	2
	Class weight	{None, balanced}	balanced
TF-IDF Feature Extraction			
Parameter			**Value**
max_features			10,000
ngram_range			(1, 2) - unigrams + bigrams
stop_words			English standard list
min_df / max_df			2 / 0.95
sublinear_tf			True (log scaling)

Table 7: Statistical Analysis (Track 1): Linear Regression and Hypothesis Testing. Paired t-tests compare 100% synthetic against 0% baseline with Benjamini-Hochberg FDR correction ($\alpha = 0.05$).

Metric	GPT-4.1-mini	Claude-3.5-Haiku
Linear Regression		
Slope (β_1)	-0.0011	-0.0039
R^2	0.982	0.976
Interpretation	-0.11% per 1% synth	-0.39% per 1% synth
Hypothesis Testing		
t-statistic	-8.32	-25.41
p-value	$< 10^{-6}$	$< 10^{-15}$
Cohen's d	-2.67 (large)	-9.18 (large)